

MSK iMCQ 07

Genetics & Histology Cell Biology

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Genetics Questions

A 2-year-old boy is brought to the physician by his parents for a routine health maintenance examination. The parents report that he held his head up at 7 months, sat without support at 11 months, began crawling at 10 months, but has not begun walking yet; findings consistent with delayed milestones. Physical examination shows distinctive craniofacial features with macrocephaly, frontal bossing, midface retrusion, and saddle nose deformity. He has disproportionate short stature with pronounced shortening of both upper and lower limbs. Which of the following best explains the pathogenetic mechanism in this patient's condition?

- | | |
|--|-----|
| A. Dominant negative effect | 20% |
| B. Gain of function mutation | 20% |
| C. Haploinsufficiency | 20% |
| D. Triplet nucleotide repeat expansion | 20% |
| E. Loss of heterozygosity | 20% |

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A 25-year-old man comes to the physician because of a 5-year history of slurred speech, vision, and hearing difficulties. His 35-year-old brother was diagnosed 15 years ago with a genetic disorder known to cause progressive abnormal mitochondrial function affecting chromosome 9. The brother is wheelchair bound and has a 1-month history of chronic heart failure. His pulse is 80/min, respirations are 14/min, and blood pressure is 120/75 mmHg. Physical examination shows decreased upper and lower limb reflexes. Laboratory studies show fasting serum glucose concentration is 150 mg/dL (Normal: 70-100 mg/dL), serum creatine kinase is 80 U/L (Normal: 25-90 U/L). Which of the following best describes this disorder?

- A. Premutation expansion in the promoter of the FMR1 gene 17%
- B. Heterozygous deletion of the FMR1 gene promoter 17%
- C. CAG repeat expansion in an exon 17%
- D. GAA repeat in an intron of Frataxin gene 17%
- E. Trinucleotide repeat expansion in 3'UTR of the DMPK gene 17%
- F. Premature stop codon in the DMPK gene 17%

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A 40-year-old man is brought to the physician by his wife because of a 1-year history of falling that has progressively worsened in frequency and severity. He also noted a progressive bilateral hand weakness with occasional cramping, which worsens in cold temperatures. He has difficulty releasing his grasp after a handshake. Physical examination shows distal upper and lower limb muscle weakness with normal proximal muscle strength. Cardiac auscultation shows normal heart sounds with a nondisplaced apex. An electrocardiogram (ECG) shows intraventricular conduction delay. Which of the following is the most likely diagnosis?

- | | |
|------------------------------|-----|
| A. Becker muscular dystrophy | 17% |
| B. Myotonic dystrophy | 17% |
| C. Osteogenesis imperfecta | 17% |
| D. Ehlers-Danlos Syndrome | 17% |
| E. Marfan Syndrome | 17% |
| F. Huntington disease | 17% |

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A 62-year-old woman comes to the physician for a follow-up examination. She was prescribed metoprolol because her blood pressure was consistently 180/110 mmHg two months ago. Her blood pressure at the current visit is 170/100 mm Hg, and she says she is adherent to the treatment. Which of the following best explains the present blood pressure findings in this patient?

- A. Variant in the CYP2C9 gene 20%
- B. Extensive metabolizer of metoprolol 20%
- C. Poor metabolizer of metoprolol 20%
- D. Amplification of the CYP2D6 gene 20%
- E. Variant of the TPMT gene 20%

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A 28-year-old man comes to the physician for genetic counseling after being diagnosed with a genetic disorder in childhood. He is planning to start a family with his partner and wishes to know the risk of his children developing the disorder. His partner has no family history of major medical illnesses. He is 200 cm (6 ft 7 in) tall and weighs 77.1 kg (170 lb). Physical examination shows long extremities, a sunken chest, and a diastolic murmur in the right parasternal region located in the 2nd intercostal space. An ophthalmic examination shows a superior and lateral subluxation of the lens. Which of the following best represents the risk of his children being affected?

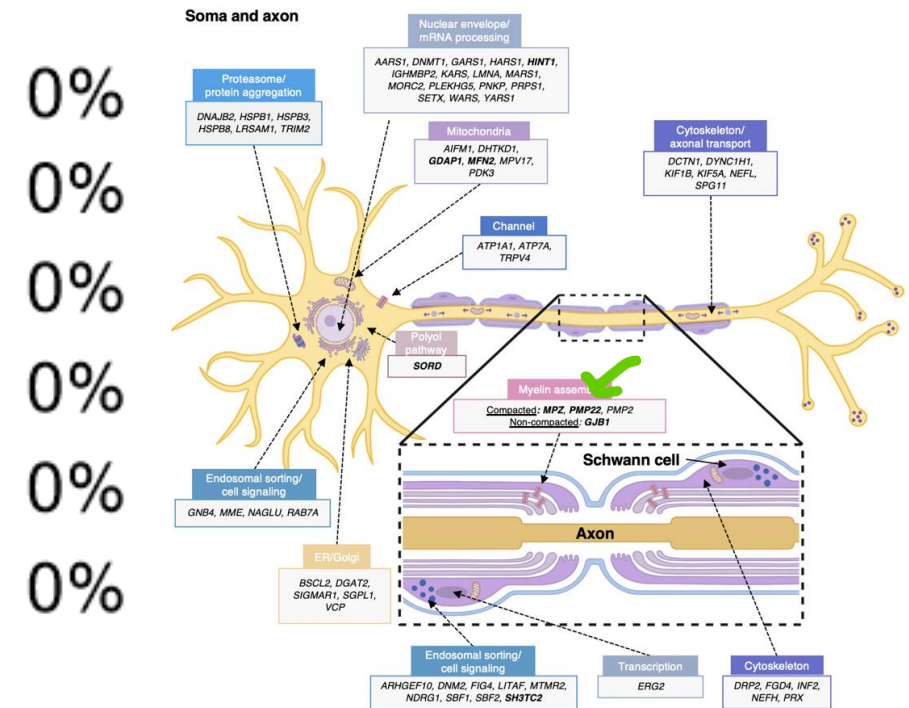
- | | |
|----------------------------|-----|
| A. 100% (all his children) | 13% |
| B. All his male children | 13% |
| C. All his daughters | 13% |
| D. 66% | 13% |
| E. 50% | 13% |
| F. 33% | 13% |
| G. 25% | 13% |
| H. None of his children | 13% |

A 28-year-old man comes to the physician for genetic counseling after being diagnosed with a genetic disorder in childhood. He is planning to start a family with his partner and wishes to know the risk of his children developing the disorder. His partner has no family history of major medical illnesses. He is 200 cm (6 ft 7 in) tall and weighs 77.1 kg (170 lb). Physical examination shows long extremities, a sunken chest, and a diastolic murmur in the right parasternal region located in the 2nd intercostal space. An ophthalmic examination shows a superior and lateral subluxation of the lens. Which of the following best represents the risk of his children being affected?

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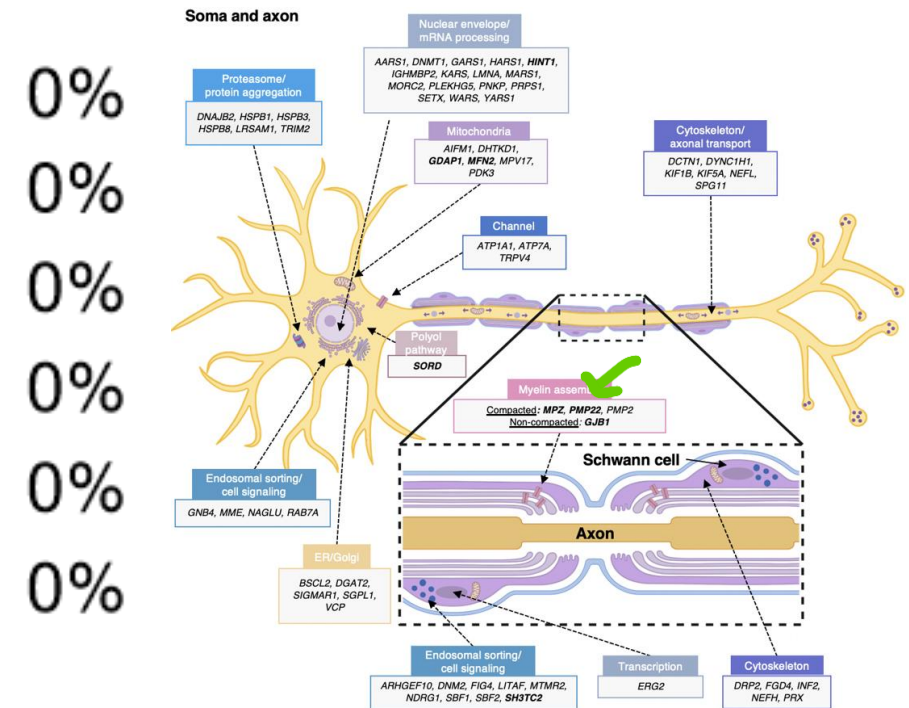
A 12-year-old boy is brought to the physician by his mother because of a 3-week history of difficulty walking. He says his toes seemed to scrape the ground while walking. His mother and maternal grandfather have a similar history, and a progressive genetic disorder has been identified. Physical examination shows high arches and weakness in hand grip strength. Genetic testing shows a pathogenic variant of the *PMP22* gene located on chromosome 17. Which of the following pathogenic variants of the gene is most likely in this patient?

- A. Microdeletion of contiguous genes
- B. Gene duplication
- C. Frame-shift mutation
- D. Triplet repeat expansion in an exon
- E. Reciprocal translocation
- F. Homozygosity for a pathogenic variant allele



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HCB Questions

A 72-year-old woman comes to the physician because of a 6-month history of dry, itchy skin on her lower legs that seems to be getting worse. She reports that it is extremely bad during the winter months. Physical examination shows dry, cracked skin with scaling as seen in the attached image. A diagnosis of age-related xerosis (dry skin) is made. Which of the following best describes a component of the epidermis most likely disrupted to result in this condition?

- A. Contains abundance of desmosomes
- B. Composed of the cell envelope and the lipid envelope
- C. Involved in the pigment melanin
- D. Contains an abundance of sebocytes for sebum production
- E. Composed of myelinated nerve endings surrounded by a capsule



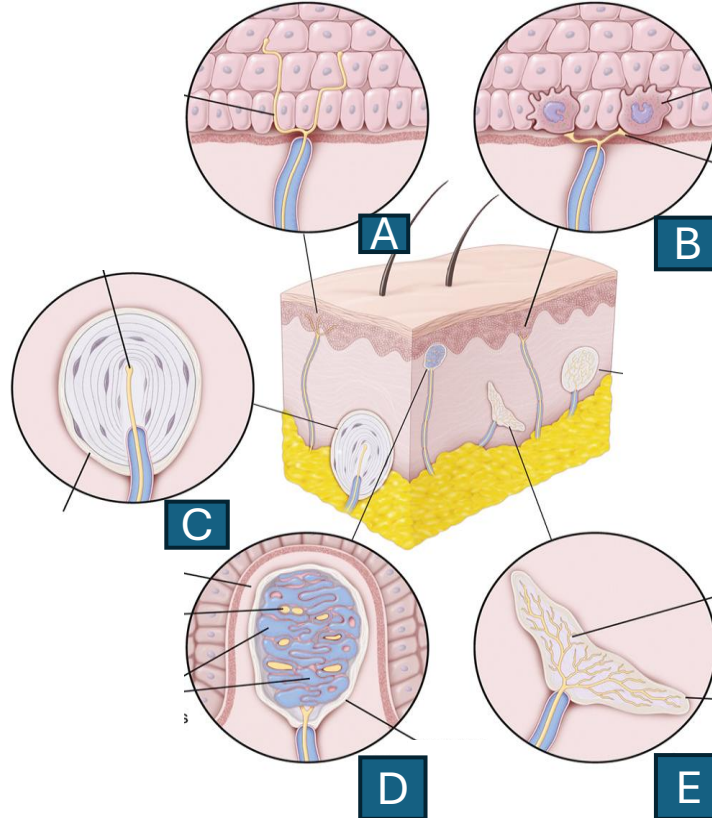
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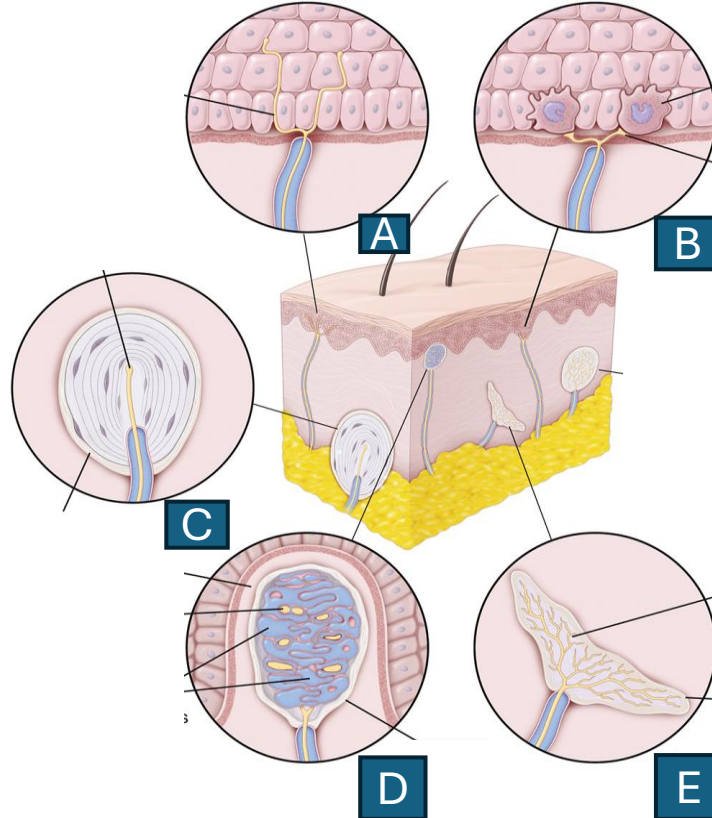
A 3-year-old boy is brought to the physician by his parents because of a 3-day history of increased redness, itching and scaling on his cheeks and forearms. His parents report that the child has been experiencing increased skin dryness and sensitivity to topical products over the past month. Physical examination shows generalized dried, scaly skin. Which of the following sensory nerve receptors in the attached image is most likely responsible for the itching sensation experienced by this patient?

- A. A
- B. B
- C. C
- D. D
- E. E



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- A. ☒ A
- B. B
- C. C
- D. D
- E. E



A 57-year-old woman comes to the physician because of a 6-month history of dry, itchy skin that seems to be getting worse. Physical examination shows regions of thickened, rough, scaly skin. Examination of a biopsy specimen from one of the affected regions shows an abnormally increased size of the area outlined by the brackets in the attached photomicrograph. Which of the following best defines this biopsy finding?

- A. Increased thickness of the stratum spinosum
- B. Excessive thickening of the stratum corneum
- C. Separation of the epidermal cells from the basement membrane
- D. Retention of nuclei of keratinocytes in the stratum corneum
- E. Keratinocyte invasion deep into the dermis



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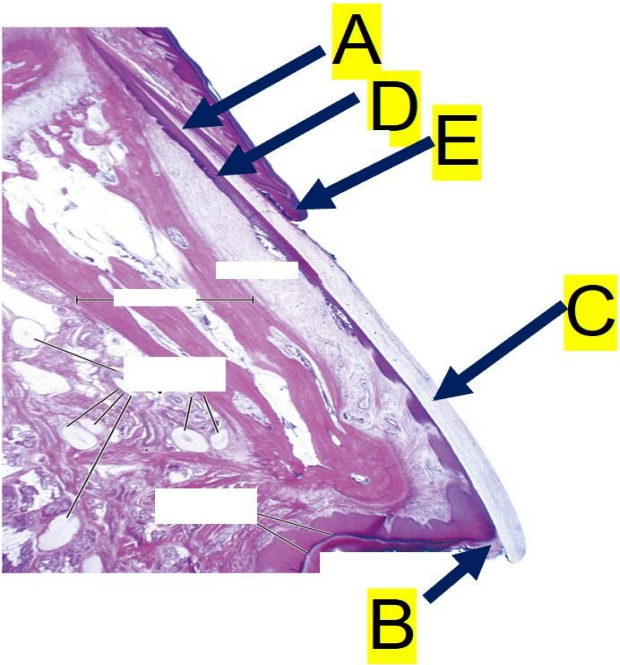


A 35-year-old woman comes to the physician because of a 3-day history of a red, swollen, tender nail fold on her finger, and slight discoloration of the nail plate as seen in the attached image labeled “A”. She had just received a manicure 1 week ago, which involved trimming her cuticles. Physical examination shows pus draining from the side of the finger (see image “A”) and increased warmth over the distal finger. The physician suspects that the eponychium is infected. Which of the following in the attached photomicrograph best identifies the infected part of the nail?

- A. A
- B. B
- C. C
- D. D
- E. E



<https://my.clevelandclinic.org/-/scassets/images/org/health/articles/15327-nail-infection.ashx>

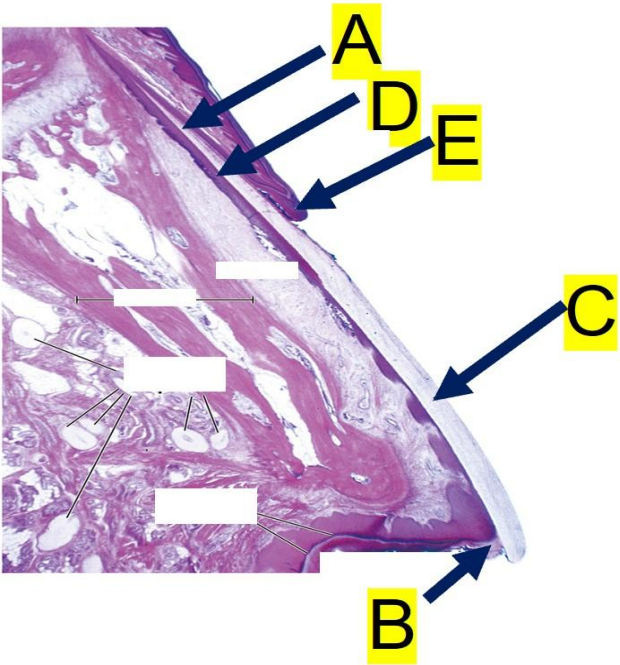


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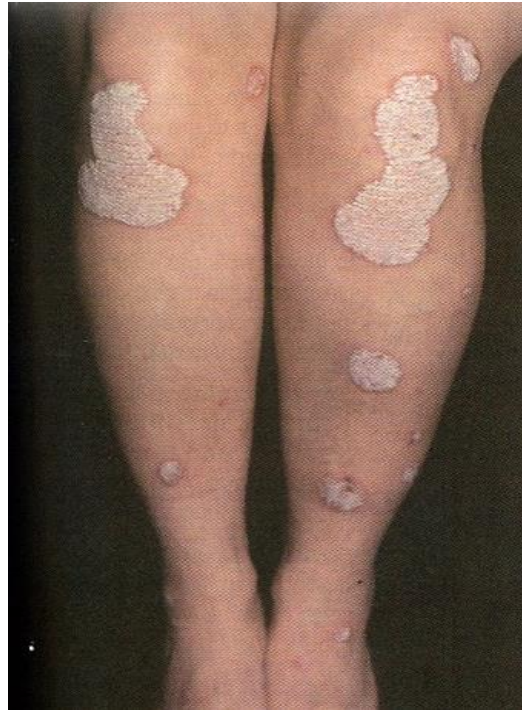
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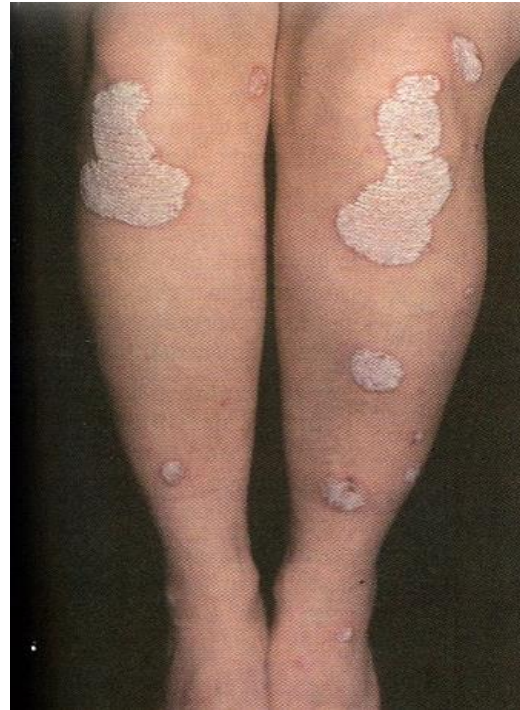


A 45-year-old man comes to the physician because of a 3-month history of itchy red, scaly patches on his elbows, knees, and scalp. Physical examination shows sharply defined erythematous plaques covered with silvery scales as seen in the attached image. Microscopic examination of a biopsy specimen taken from one of the plaques show hyperkeratosis, parakeratosis and Munro's micro abscesses. A diagnosis of psoriasis is made. Which of the following ranges best characterizes the normal epidermal turnover time?



- A. 8 – 10 days
- B. 1 – 2 days
- C. 1 – 47 days
- D. 10 – 12 days
- E. 14 – 26 days

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A 68-year-old man comes to the physician because of an 18-month history of a slow-growing, non-healing sore on his nose that occasionally bleeds. He has a history of significant sun exposure due to his job. Physical examination shows a fair-skinned male with a pink, pearly papule with rolled borders, and small dilated blood vessels (telangiectasias) on the side of his nose, as seen in the attached image. Examination of a biopsy specimen of the papule shows nests of basaloid cells with a characteristic palisading appearance of the cells at the periphery of the lesion, and invasion into the dermis. A diagnosis of a malignancy is made. Which of the following cell types is most likely affected in this condition?

- A. Squamous
- B. Melanocytes
- C. Langerhan
- D. Basal
- E. Merkel



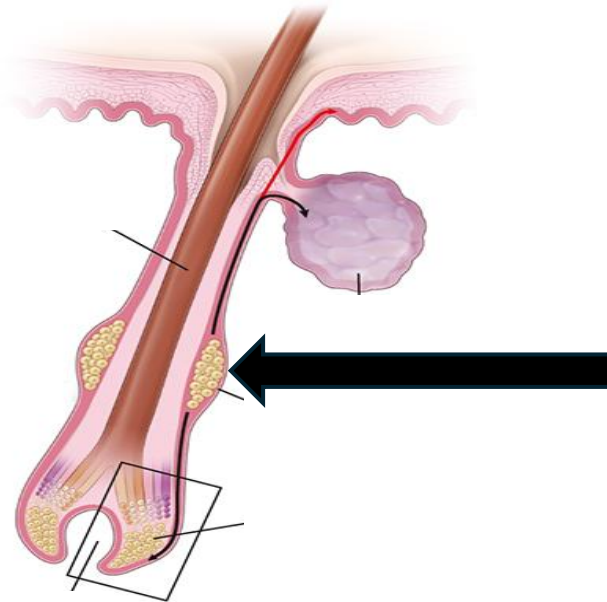
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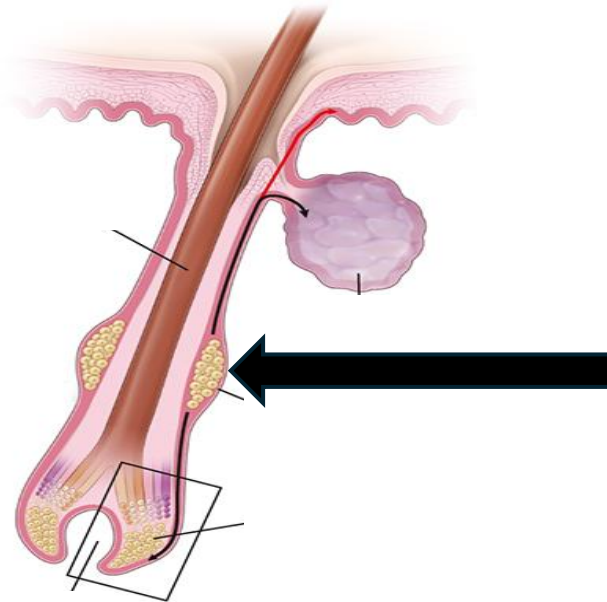
A 28-year-old man comes to the clinic because of a 2-wk history of multiple, painful, erythematous papules and pustules on his buttocks and thighs. He works as a restaurant cook and has a history of wearing tight-fitting, occlusive clothing for long hours. Physical examination shows redness and swelling around the hairs of the skin of the affected areas. Microscopic examination of a biopsy specimen from one of the swelling shows neutrophil infiltration at the area shown at the tip of the arrow in the attached image. Which of the following best identifies this area?

- A. Follicular bulge
- B. Inferior segment
- C. Hair shaft
- D. Isthmus
- E. Infundibulum



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A 30-year-old man comes to the physician because of a 2-week history of small, itchy tender, pus-filled bumps on his neck and scalp, that started after shaving. Examination of a biopsy specimen of one of the bumps show a neutrophilic infiltrate of the inferior segment of the hair follicle, with involuted hair matrix cells. Which stage of hair growth is primarily affected in this patient?

- A. Anagen
- B. Catagen
- C. Telogen

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A 45-year-old man comes to the emergency department because of a 1-hour history of a deep laceration on his lower leg following an accident. His vital signs are within normal limits. Physical examination show a 7-cm lesion involving multiple layers of skin, and a blood clot in the lesion. Which of the following is most likely a microscopic finding at this stage of deep wound healing?

- A. Vasodilation and immune cell activation
- B. Platelet aggregation and activation of the coagulation cascade
- C. Vasoconstriction and vasodilation
- D. Angiogenesis and epithelialization
- E. Collagen remodeling and scar formation

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A 50-year-old woman comes to the physician because of a 4-month history of painful, non-itchy blisters in her mouth that have recently progressed to affect her scalp, chest, and back. These blisters rupture easily, leaving behind raw, painful erosions. The physician suspects pemphigus vulgaris. Which of the following microscopic findings best characterizes this condition?

- A. Separation of keratinocytes from the basement membrane
- B. Hypertrophy of sebaceous glands
- C. Separation of keratinocytes in stratum spinosum
- D. Hyperkeratosis and parakeratosis
- E. Parakeratosis alone

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